

Bang for the Buck: Modeling NBA Player Performance to Contract Value

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Background

Professional sports sit at the intersection of **athletic competition and business management**. In the NBA, teams must build rosters capable of winning within a hard salary cap, which makes every contract decision consequential.

Player salaries have grown dramatically alongside **rising cap values** over the past decade. Despite this, “overpaid” and “underpaid” labels remain largely **opinion-based**. Advanced statistics are now **widely tracked and publicly available**, making the NBA an ideal setting for analyzing how well contracts actually reflect on-court value.

Yet no consistent, data-driven framework exists for evaluating whether a player’s pay matches their production. **This project sets out to build one**. Using performance metrics across ten seasons (2016-17 through 2025-26), we model expected salary as a percentage of the cap to **identify over- and undervalued players** and examine whether **younger players are more systematically underpaid**. We also explore whether **contract timing shapes production**, building on evidence that performance dips the season after signing even when the contract year itself shows no significant change (Owsik & Tan, 2026).



Motivation

Contract negotiations pit **players seeking market value** against **front offices operating under a fixed budget**, a tension that inevitably produces mismatches. As salaries and cap values have grown, so too has the stakes of getting these decisions right. Some players earn more than their production justifies, while others earn far less.

Despite constant debate among fans and analysts, “overpaid” and “underpaid” remain largely subjective labels with no consistent framework behind them. This project applies a **data-driven approach** to move beyond the vibes and quantify these gaps systematically across a decade of NBA data.

Guiding Questions

1. Can a player’s “fair” salary be estimated from their stats, and who looks most over- or undervalued?
2. Does age affect how accurately the model predicts salary? Are younger players more systematically underpaid?
3. Does contract timing affect performance? Do players produce more before signing and less after?

Data Sources: NBA API & Basketball Reference

Primary Dataset: Player Performance Statistics

Player performance statistics form the foundation of our salary prediction model. We retrieved advanced metrics using the `nba_api` Python library, focusing on efficiency and context-adjusted measures rather than raw counting stats. Advanced metrics such as PIE (Player Impact Estimate), net rating (point differential per 100 possessions), and usage rate (share of team possessions used while on floor) capture a player’s true contribution to winning and are stronger predictors of contract value than points, rebounds, or assists alone.

We called the `LeagueDashPlayerStats` endpoint across all 10 target seasons for regular season games, retrieving advanced measures for each per player per season. Playoff records were excluded to prevent postseason appearances from skewing the analysis. A 3-second sleep between requests avoided placing excessive load on the NBA’s statistics server.

Location	stats.nba.com via <code>nba_api</code> (<code>LeagueDashPlayerStats</code>)
Format	JSON → Pandas DataFrame
Variables	PTS, REB, AST, PIE, TS_PCT, USG_PCT
Records	5,492 records (81 columns x 5,492 rows)
Period	2016-17 through 2025-26

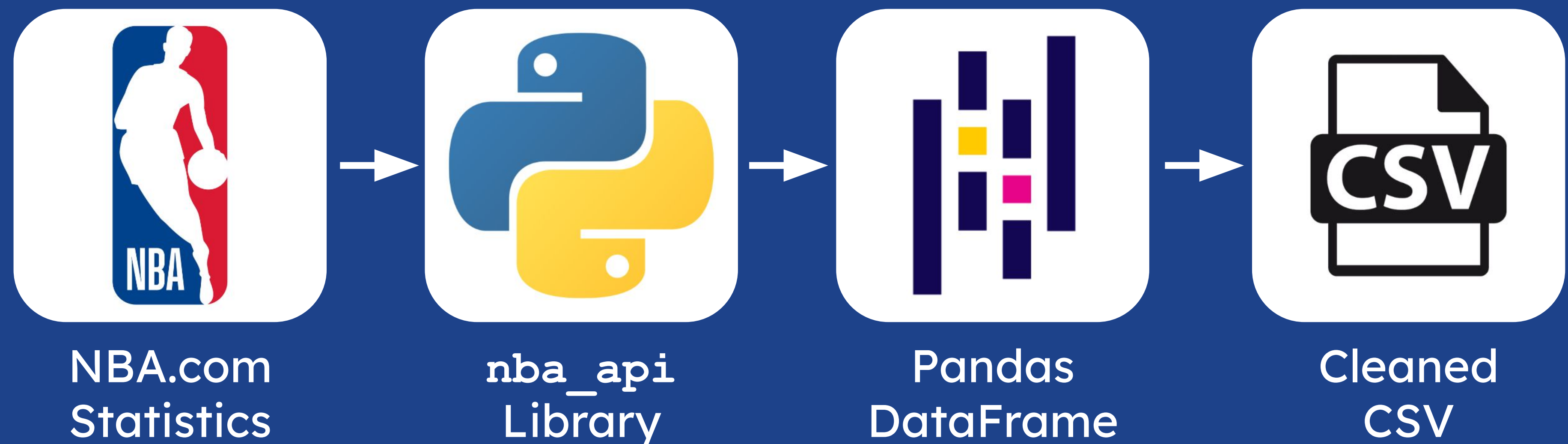
Secondary Dataset: Player Salary Data

Salary data provides the target variable for our model. We collected each player’s annual salary alongside the season’s cap value, allowing us to normalize pay across seasons and make meaningful comparisons year over year. Players earning less than 1% of the salary cap were excluded, as two-way and 10-day contracts are not relevant to salary bump analysis. Basketball Reference publishes this data at the team level, so we retrieved a salary page for each of the 30 NBA teams across all 10 target seasons.

We scraped each team’s salary page using Python’s `requests` library and `BeautifulSoup`, totaling 300 sequential requests. A key technical challenge was that Basketball Reference embeds salary tables inside HTML comments, requiring a two-pass parsing approach to extract them. A randomized 1–3 second delay between requests prevented the HTTP 429 rate-limit errors we encountered in earlier runs.

Location	<code>basketball-reference.com/teams/{TEAM}/{YEAR}.html</code>
Format	HTML (scraped) → Pandas DataFrame
Variables	player, team, season, salary
Records	6,535 records (14 columns x 6,535 rows)
Period	2016-17 through 2025-26

Data Collection/Manipulation: Player Performance Statistics (nba_api)



Data Collection

Player statistics were retrieved from NBA.com via the `nba_api` library's `LeagueDashPlayerStats` endpoint across 10 seasons (2016-17 through 2025-26). The Advanced measure type was used, providing efficiency and context-adjusted metrics such as PIE, usage rate, and net rating that are stronger predictors of contract value than counting stats alone. Each season was requested separately and tagged with a `SEASON` column before being concatenated into a single dataframe. A 3-second sleep between requests avoided placing excessive load on the NBA stats server.¹

Data Cleaning

Playoff records were filtered out to ensure statistics reflect regular season performance only, preventing playoff appearances from skewing the analysis. Players appearing on multiple teams within a season had their `TEAM_ABBREVIATION` set to `TOT` to match the convention used in the salary dataset.² A normalized join key in the format (e.g., `lebr_james_1a1_24`) was then constructed to enable merging with the salary data.

Data Manipulation

`TEAM_COUNT` was computed by grouping records by player and season before any aggregation, capturing how many teams each player appeared on during the year. This count was preserved as a flag for multi-team players and used to drive the `TOT` assignment step in cleaning.

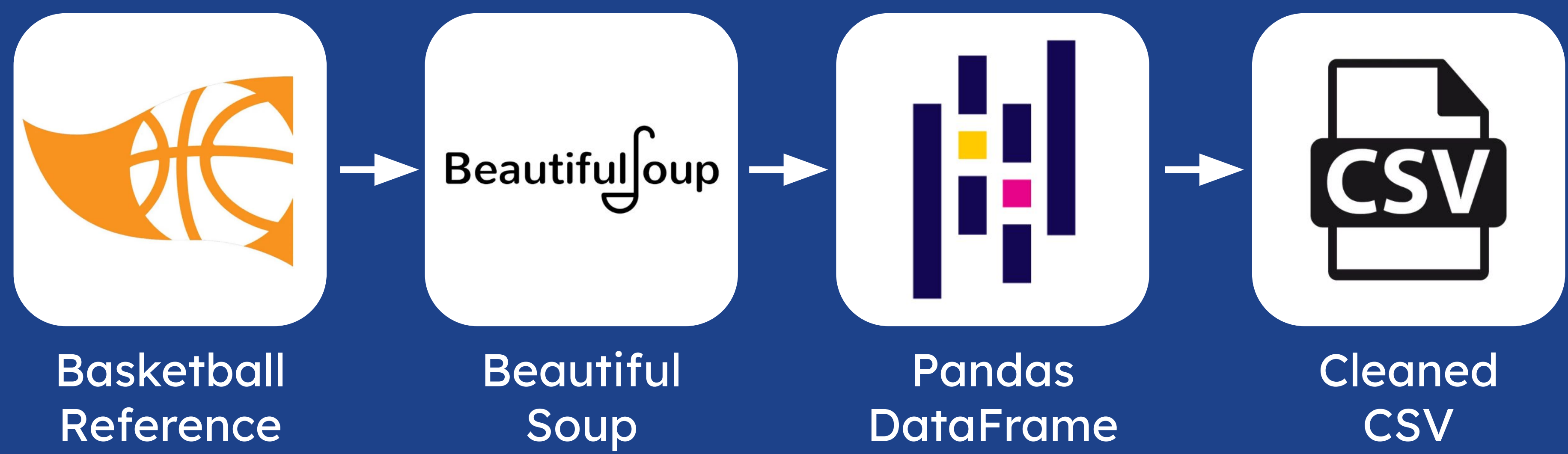
A subset of 19 analytically relevant columns was retained from the 81 returned by the API, dropping per-stat ranking columns that reflect league-wide rank rather than the underlying metric. The retained columns cover efficiency metrics, playing time, win-loss context, and context-adjusted ratings, giving the model a comprehensive view of each player's contribution without introducing redundant features. The final dataset contains 4,450 records and was exported to CSV for use in the join pipeline.

Technical Challenges

¹ The NBA stats server occasionally returns timeout errors under load. A fixed 3-second sleep between requests kept the pipeline stable across all 10 sequential calls. A 60-second timeout parameter was also set on each request to prevent the pipeline from hanging indefinitely on a failed response.

² Players on multiple teams appear as separate per-team records in the raw API response. `TEAM_COUNT` had to be derived before setting `TEAM_ABBREVIATION` to `TOT`, as the aggregation step collapses those records into one and the count would otherwise be lost.

Data Collection/Manipulation: Player Salary Data (Basketball Reference)



Data Collection

Salary data was scraped from Basketball Reference team pages across all 30 teams and 10 seasons (2016-17 through 2025-26), totaling 300 sequential requests. Each page’s salary table is embedded within HTML comments rather than standard HTML, requiring a two-step BeautifulSoup parsing approach to locate and extract the table correctly.¹ A randomized 1-3 second delay was added between requests to avoid triggering HTTP 429 rate-limit errors encountered in earlier runs.

Data Cleaning

During cleaning, UTF-8 encoding was enforced on all responses to handle international player names (e.g., Luka Dončić). All scraped fields were converted to correct data types. Players earning less than 1% of the salary cap were filtered out, as two-way and 10-day contracts are not relevant to salary bump analysis.

Players who appeared on multiple teams within a season had their salaries aggregated into a single record with team set to TOT.²

Data Manipulation

Once cleaned, additional features were engineered to support salary normalization and contract analysis. `salary_cap`, `salary_cap_pct`, and `salary_rank` were added so salaries could be compared meaningfully across seasons as the league cap has risen over time. `salary_tier` categorized each player into one of four cap percentage bands (0-8%, 8-15%, 15-25%, 25%+) to group players by market value, providing a useful lens for segmenting the analysis by contract level.

The dataset was then sorted by player and year before deriving cross-season features.³ `salary_pct_change` measures year-over-year salary movement, while additional flags capture meaningful contract events such as contract years, max deals, and mid-season team changes. A join key in the same format as the stats dataset was constructed to enable merging. The final dataset contains 4,420 records and was exported to CSV.

Technical Challenges

- ¹ Initial scrape attempts returned empty dataframes. A second parse pass using BeautifulSoup’s `Comment` class was required to locate the tables embedded in HTML comments.
- ² Records were grouped by player and year, with salaries summed and team set to TOT for multi-team players before creating `salary_rank`.
- ³ Sorted by player and year first, as `.shift()` compares against the prior season and incorrect ordering produces false positives.

Combined Dataset Summary & Initial EDA

Joining the Datasets

The two datasets were joined on a shared `id` key in the format `{first_four_letters_of_first_name}_{last_name}_{team}_{year}` using an inner join, producing 3,268 matched records from 4,450 stats records and 4,420 salary records. Applying the analysis thresholds of `salary_cap_pct >= 1.0%` and `total_minutes >= 500` narrowed this to 2,973 player-seasons used in the analysis.

Unmatched Records

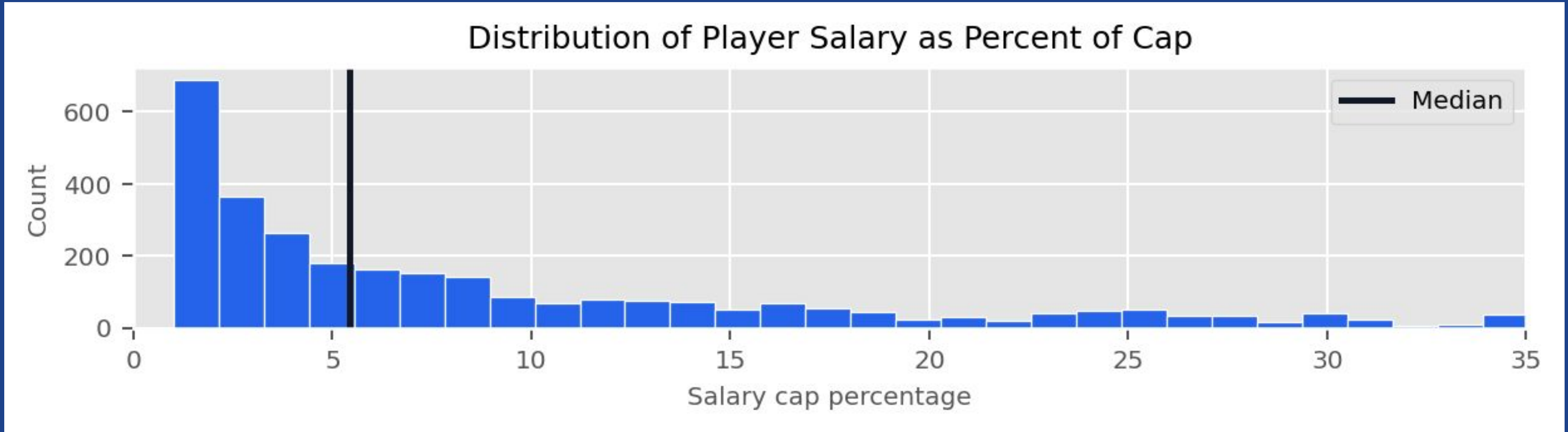
1,152 salary records and 1,182 stats records did not join. Two primary patterns explain the majority of mismatches:

- TOT alignment gap.** NBA.com lists a single aggregated record under `TOT` for players who appeared on multiple teams, while Basketball Reference lists each team stint separately, causing the same player-season to carry different team codes (e.g., `aaro_gordon_tot_21` vs. `aaro_gordon_den_21`).
- Name formatting inconsistencies.** Players with initials such as A.J. or J.J. were formatted differently across sources (e.g., `aj_green` vs. `a.j._green`).

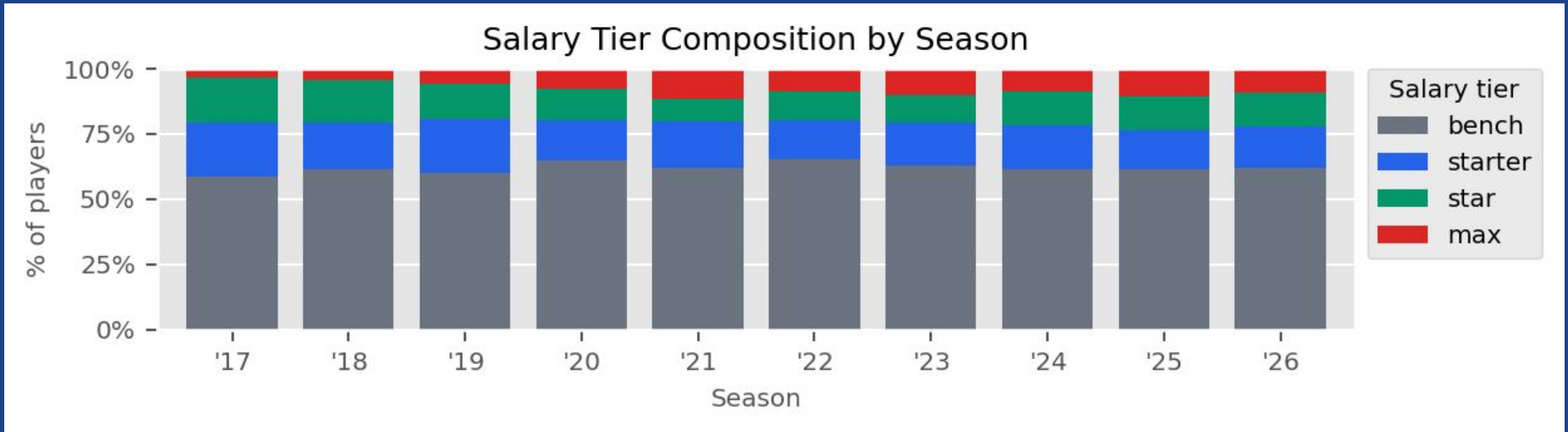
The remaining unmatched records reflect genuine gaps between the two sources, as the thresholds were designed to retain only players with significant playing time and meaningful contract value. If this analysis were extended, revisiting the join logic to normalize `TOT` assignment and name formatting across both pipelines would recover a meaningful portion of these records.

Initial Exploratory Data Analysis

The joined dataset contains 2,973 player-season records spanning the past ten seasons. Salary distribution is heavily right-skewed, with a median of 5.15% of the cap and most records concentrated below 10%.



Bench-tier players account for an average of 63.4% each season, while max-tier players represent just 7.4% on average, reflecting the natural pyramid structure of NBA rosters. Tier composition has remained relatively stable across the decade, though max-tier representation nearly tripled from 3.2% in 2016-17 to a peak of 10.7% in 2020-21 during the supermax contract era.



What drives NBA salary?

Volume stats outpace efficiency metrics as the strongest predictors of salary cap percentage

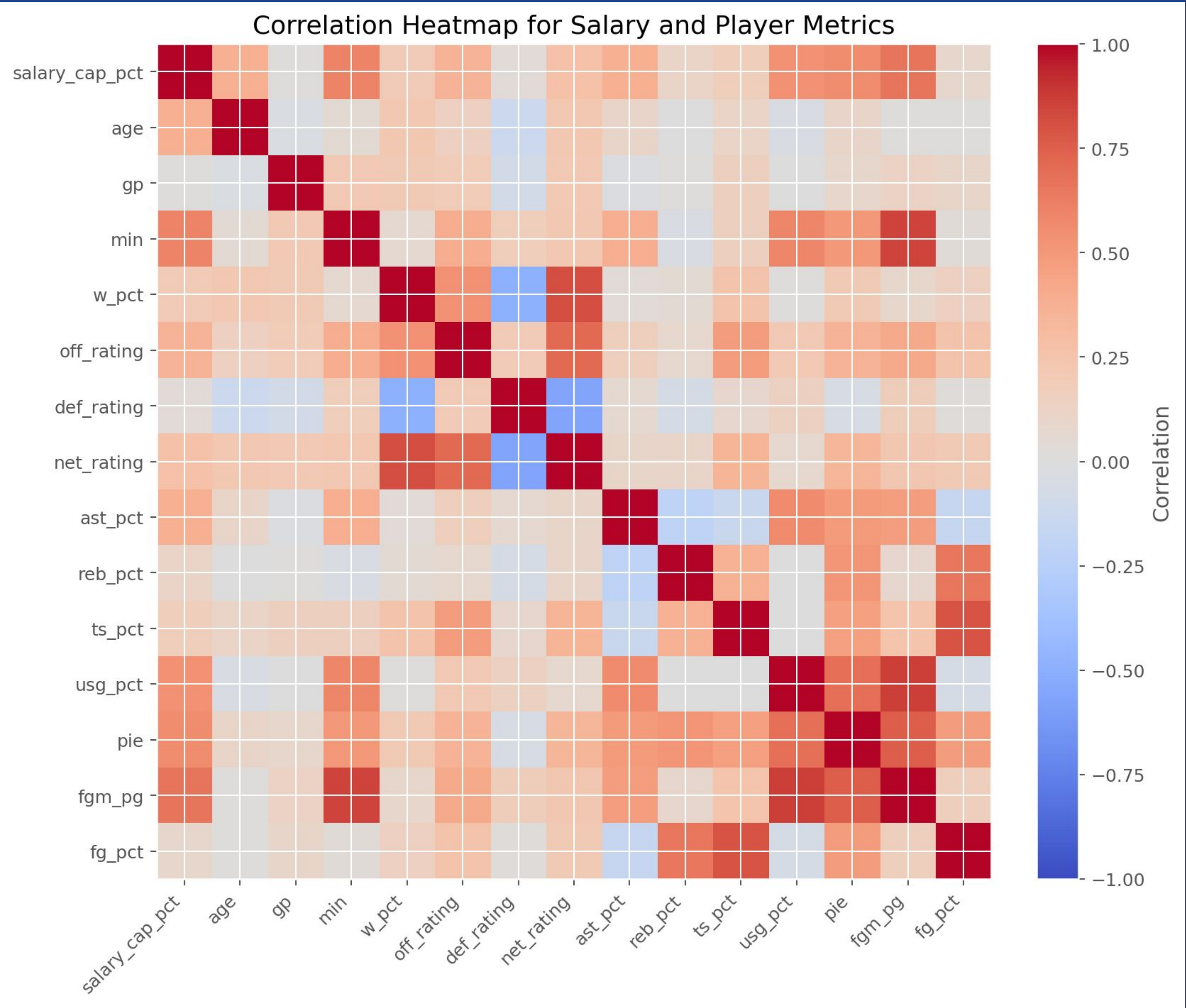
The four features most correlated with `salary_cap_pct` are `fgm_pg` (0.657), `min` (0.596), `pie` (0.555), and `usg_pct` (0.537). `salary_cap_pct` expresses each player's salary as a percentage of that season's cap, normalizing pay across seasons as the league cap has risen over time. Together the four features capture **two distinct signals: volume** (`fgm_pg`, `min`, `usg_pct`) **and overall impact** (`pie`).

The dominance of volume stats is consistent with prior research. Arkow et al. (2023) found that "traditional stats like points or blocks tend to be overvalued, while analytical or efficiency stats like eFG%, Win Shares, or PER are undervalued," suggesting the **market rewards scoring involvement more heavily than overall impact**. This pattern is visible in our own correlations, where **`fgm_pg` outpaces `pie` as the strongest salary predictor**, suggesting efficient but low-usage players may be systematically undervalued while high-volume scorers earn a premium their efficiency numbers do not fully justify.

Two pairs of collinear features were identified in the joined dataset. `min` and `total_minutes` capture the same playing time signal, and `fgm_pg` and `fga_pg` are structurally linked by shot volume. Only `min` and `fgm_pg` were retained for further analysis.

Age also shows a moderate positive correlation with `salary_cap_pct`, reflecting the structural step-up players experience when moving from rookie contracts to open market deals. Its relationship with salary is explored further in the analysis of systematic underpayment by age.

These four features carry through the rest of the report as the foundation for estimating fair salary and identifying where the market appears to incorrectly value players.



Top Correlations with Salary Cap Percentage

- 1. `fgm_pg` (Field Goals Made Per Game): 0.657
- 2. `mins` (Minutes): 0.596
- 3. `pie` (Player Impact Estimate): 0.555
- 4. `usg_pct` (Usage Percent): 0.537

Can a player’s “fair” salary be estimated from their stats?

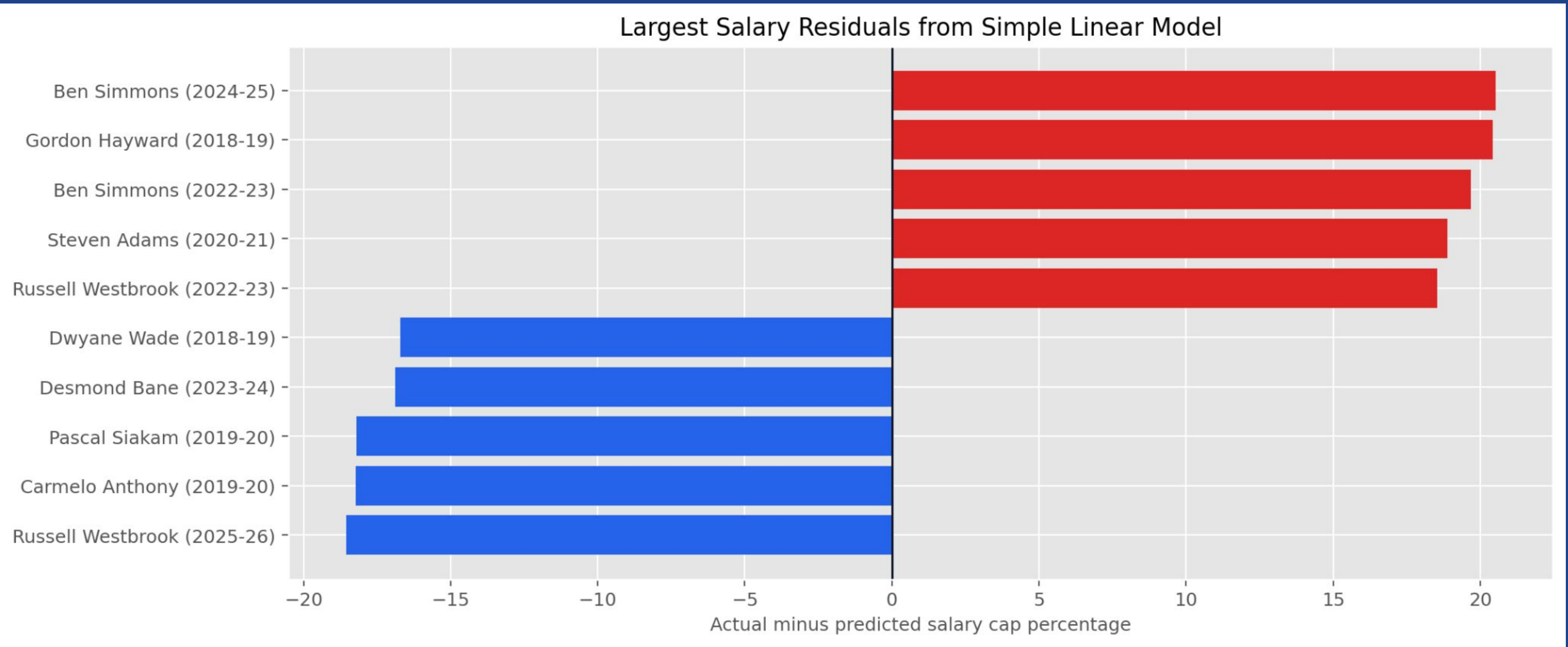
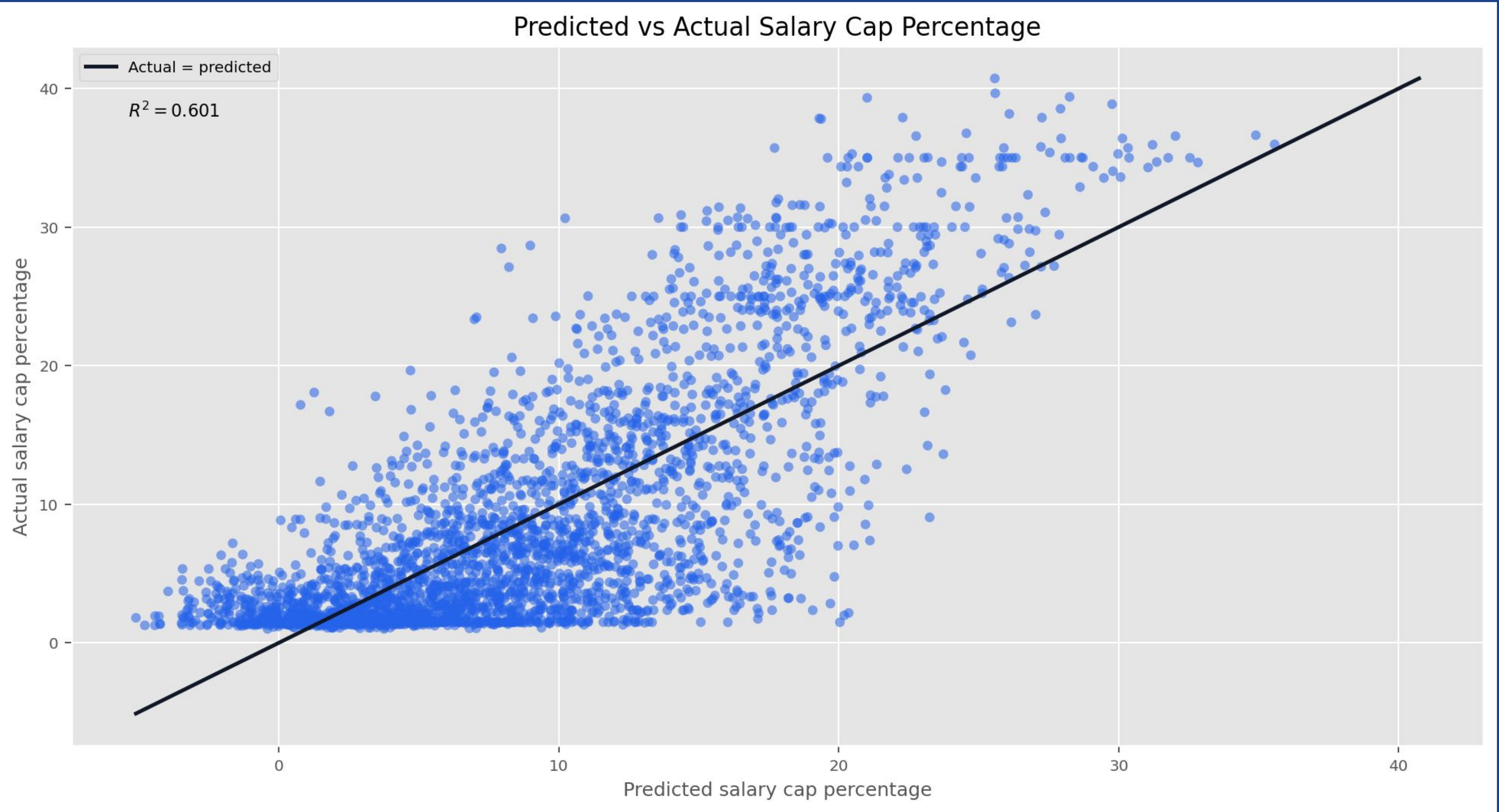
Performance metrics explain ~60% of salary cap percentage variance, but who looks most over/under-valued?

Using the features identified in the correlation analysis, we fit an OLS regression to predict salary_cap_pct from player performance metrics. The model explains **roughly 60% of salary variance**, leaving a meaningful residual that captures the gap between what a player earns and what their stats suggest they should. Players with the largest positive residuals are earning well above their production, while those with the largest negative residuals are being paid well below it.

The residuals reflect patterns on both ends. High-volume scorers on aging contracts earned well above what their declining production justified, with Ben Simmons appearing twice among the most overpaid across 2022-23 (+19.69%) and 2024-25 (+20.51%), and Gordon Hayward 2018-19 (+20.41%) as another prominent case.

Career trajectory cuts both ways, as **Russell Westbrook ranked fifth most overpaid in 2022-23 (+18.54%) on a max contract before appearing as the most underpaid player in 2025-26 (-18.56%) on a minimum deal**, a shift that reflects career decline rather than improved market efficiency. Carmelo Anthony followed a similar arc, appearing among the most underpaid in 2019-20 (-18.22%) after previously earning a max contract.

A third pattern captures structural underpayment. A third pattern captures structural underpayment. Pascal Siakam 2019-20 (-18.21%) and Desmond Bane 2023-24 (-16.87%) both appear among the most underpaid despite strong production, as **rookie contract ceilings limit pay regardless of performance**. In these cases, the residual reflects a structural feature of the CBA rather than a market mispricing.



Are younger players more systematically underpaid by the market?

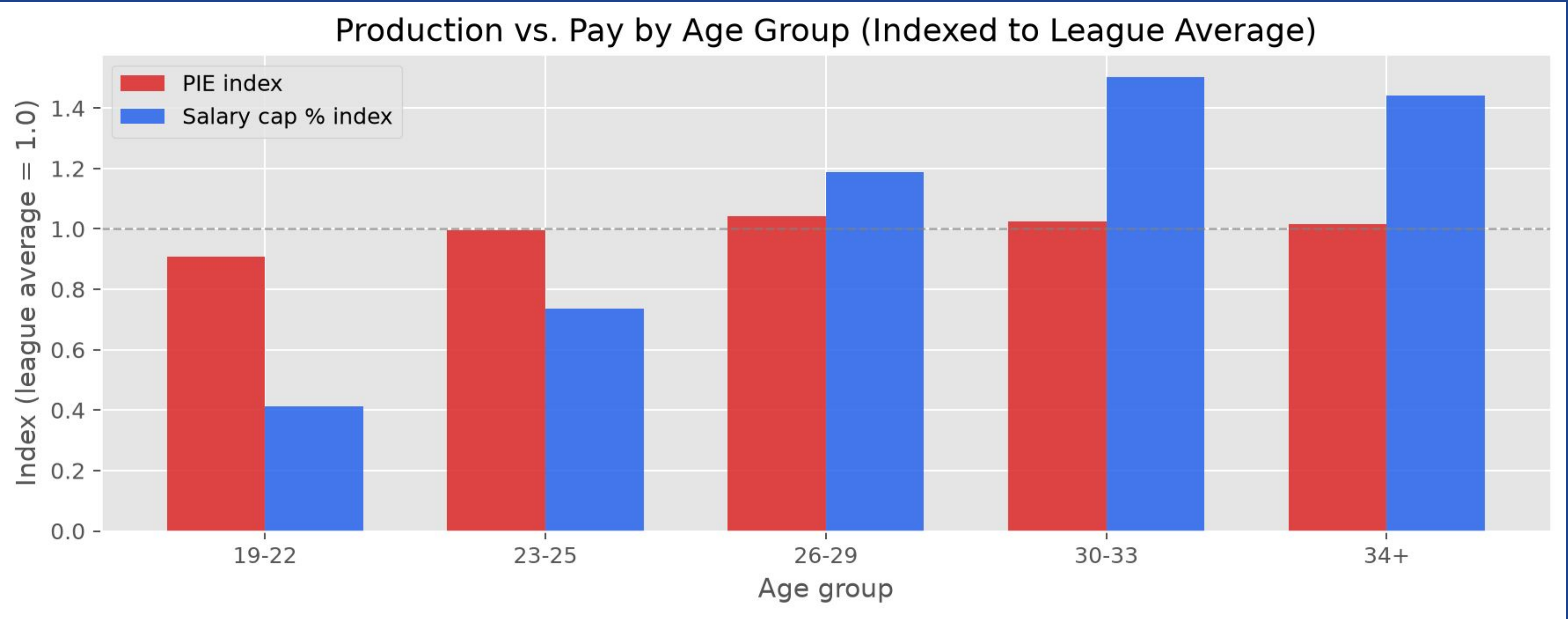
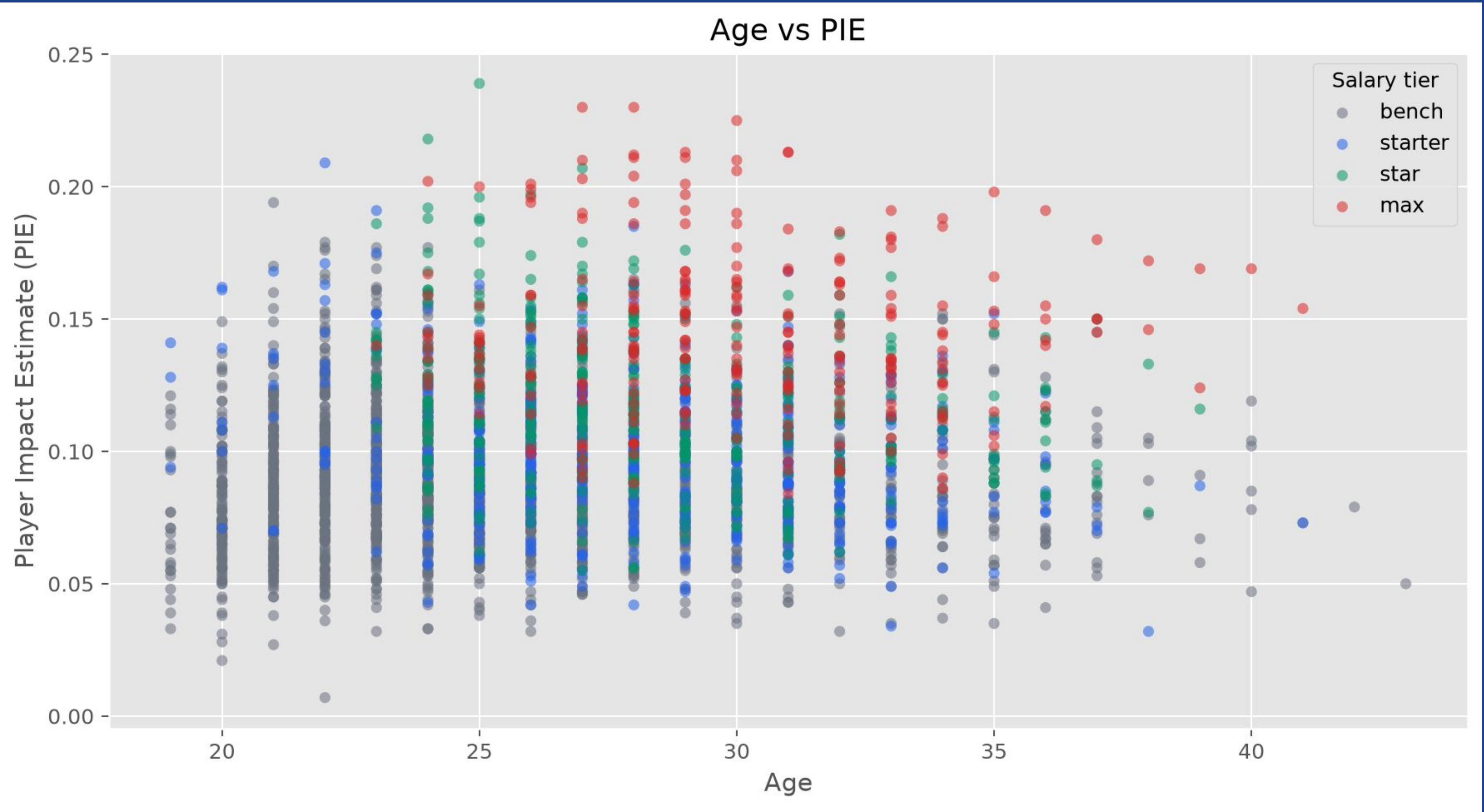
Rookie-scale contracts create a visible gap between measured impact and salary cap share

The scatterplot to the right shows PIE (Player Impact Estimate) rising through the late 20s before tapering off, while max-tier salaries cluster heavily in the 27-31 age range regardless of whether PIE has started to decline. Bench tier players (8-15% of the cap) dominate the youngest ages even when their production overlaps with starters and stars, hinting at the structural underpayment explained by age group.

The grouped bar chart below it quantifies this gap. **Players aged 19-22 post a PIE index of 0.91 (league average equals 1.0) but a salary cap % index of just 0.41**, producing near league average impact while earning less than half the league average pay share. This flips with the age group of 30-33 (1.50 salary vs. 1.02 PIE) and persists into ages 34 and older (1.44 vs. 1.02).

This matters for the salary model since it is primarily trained on performance stats like usage rate and net rating. A young player with strong PIE should be predicted to earn closer to their production level, but **rookie scale caps keep actual pay well below that, likely producing negative residuals concentrated in the 19-22 age group**. For older players, positive residuals may reflect pay for past production rather than their current value.

This pattern isn't new. Salary has historically peaked 2.5 to 3.1 years after performance peaks, with **second through fourth year players earning roughly half their market value while veterans earn roughly three times as much for comparable performance** (Rowe, 2016). This suggests our model's residuals may largely reflect this well documented structural pay lag rather than pure prediction error.



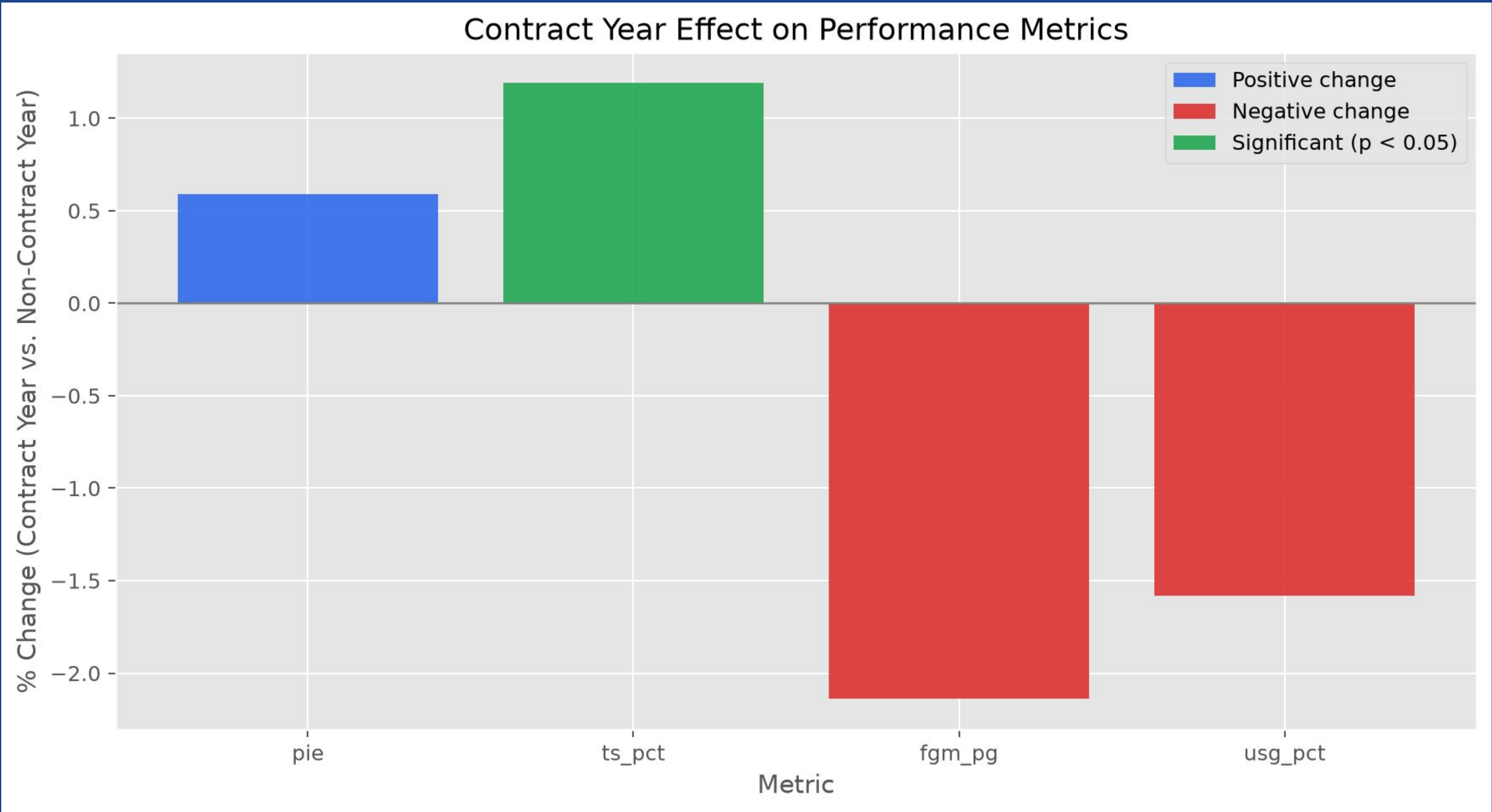
Do players produce more in contract years and less after signing?

We compared key performance metrics across contract and non-contract seasons

The bar chart to the right shows the percent change in four performance metrics from non-contract years to contract years. **True shooting percentage rose by 1.19% and is the only statistically significant change (p = 0.004)**, suggesting players shoot more efficiently in the year before signing a new contract. PIE moved only slightly (+0.59%, p = 0.7031), while field goals made per game (-2.14%) and usage rate (-1.58%) both declined, though neither reaches statistical significance.

This pattern is broadly consistent with prior research, which found that for certain subsets of players, shooting percentage, usage rate, and field goal attempts decrease in the presence of a contract year (Gaffaney, 2013). Our usage rate and field goal results align with this while our TS% result runs the opposite way, possibly reflecting players prioritizing shot quality and efficiency over volume in the year before a new contract.

Overall, **the evidence for a broad contract year performance boost is weak across most metrics**. The one significant result, a shooting efficiency bump, hints that players may be more selective about shot quality rather than playing more minutes or taking on a larger role. This is worth keeping in mind when interpreting salary residuals, since contract year status may explain small efficiency shifts but isn't a major driver of overall production. Taken together with the earlier findings on age and salary efficiency, this suggests **the league's pay structure is shaped more by long-term contract terms (rookie scales, max deals, aging veteran contracts) than by short-term performance swings tied to free agency timing**.



Metric	% Difference	P-Value	Significant
pie	+ 0.59%	0.7031	No
ts_pct	+ 1.19%	0.004	Yes
fgm_pg	- 2.14%	0.3589	No
usg_pct	- 1.58%	0.2536	No

Ethics Considerations & Further Research Opportunities

Ethics Considerations

Labeling players. Any model-based label of “overpaid” or “underpaid” is only as complete as the features used to build it. Our model does not capture injury history, locker room leadership, or fit within a team’s system, all of which legitimately affect contract value. We frame residuals as “given the stats we measured, the model expected X” rather than as a verdict on the player.

Causation vs. correlation. Observed differences in production around contract events can reflect many things, including age, role change, team context, or injury luck, rather than a direct behavioral response to contract incentives. We use descriptive language throughout rather than causal claims, and note the confounders we have not controlled for rather than overstating what the analysis can establish.

Contract year flag approximation. The `is_contract_year` flag is derived from year-over-year changes rather than directly observed contract signing dates or terms. Players who re-sign at similar salary levels or sign unusual contract structures may be miscategorized. Findings should be interpreted in the context of this approximation rather than as conclusions based on explicit contract classifications.

Max contract structural bias. The NBA’s collective bargaining agreement caps individual salaries regardless of performance, meaning max-contract players cannot earn more even if their production justifies it. Our model treats max and non-max contracts identically, introducing a known bias at the top of the salary distribution that should be kept in mind when interpreting the results.

Further Research Opportunities

Winning outcomes. A natural extension is testing whether teams with more underpaid or high-surplus players translate that advantage into wins, deeper playoff runs, or championship appearances. This would connect player-level residuals directly to team-level success and explore whether teams win by spending more or by spending more efficiently.

Contract structure. Replacing the salary-jump proxy with actual contract signing dates, deal length, rookie-scale status, extensions, and max-contract categories would address one of the core limitations of the project. A more precise contract structure would allow for cleaner classification of contract-year and post-extension seasons and reduce the miscategorization risk noted earlier.

Team value efficiency. Aggregating player residuals to the team level would allow front offices to be ranked by total surplus value, identifying which teams consistently receive more production per salary-cap dollar. This builds directly on the residuals already computed and produces a concrete, interpretable deliverable for team-based analysis.

Machine learning models. Comparison the OLS regression against random forest, gradient boosting, or regularized regression models would test whether non-linear relationships better predict salary and identify over or underpaid players more accurately. This directly addresses the linearity assumption underlying the current model and could meaningfully improve on the current R^2 of roughly 0.60.

Statement of Work & References

Statement of Work

Team, Project Management & Operations:

Quentinn set the initial project direction and core thesis, while Ryan managed project timelines and operations. All three members applied their domain knowledge to strengthen the analysis.

Project Proposal & Prior Research Review:

Ryan and Yuki conducted the background research and prior literature review, with Ryan overseeing the initial draft of the report.

Code & Presentation:

Ryan collected the salary dataset and led the initial data cleaning and joining process, then assisted in refining code for visualizations within the report. Quentinn handled data joining, initial analysis, and visualization development, supporting additional analyses and related visuals. Yuki managed statistical data collection, refined the notebook code, and contributed to additional analysis-related coding.

Assessment:

The team’s combined efforts supported a comprehensive approach to data processing, analysis, and statistical evaluation. Ryan and Quentinn refined the final report and presentation, while Yuki strengthened the statistical analysis, supporting code, and methodological rigor.

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